

SYSTEMIC LUPUS ERYTHEMATOSUS (SLE - SYSTEMIC LUPUS ERYTHEMATOSUS) NCLEX OVERVIEW

DEFINITION

Systemic Lupus Erythematosus (SLE) or Lupus is a chronic (long-lasting) autoimmune (immune system attacks own body) condition causing severe inflammation and potential permanent damage to organs, tissues, and joints.

PATHOPHYSIOLOGY (APOPTOSIS & AUTOIMMUNITY EXPLAINED)

Apoptosis (programmed cell death): Damaged cells shrink and form apoptotic bodies.

Phagocytosis (cell ingestion by macrophages): Normally, macrophages clear apoptotic bodies quickly.

In Lupus, ineffective phagocytosis leaves apoptotic bodies unremoved.

Leaked nuclear contents (antigens - substances that trigger immune response) trigger formation of antinuclear antibodies (ANA - antibodies attacking cell nuclei).

Antigens + antibodies form immune complexes (antigen-antibody clumps), causing inflammation and complement cascade activation (C3, C4 - proteins involved in inflammation).

Chronic inflammation damages organs permanently.

AFFECTED BODY SYSTEMS & SYMPTOMS

Lupus symptoms vary; inflammation severity ranges from mild to severe:

System	Manifestations
Skin	Butterfly rash (malar rash - butterfly-shaped rash on face), discoid rash (coin-shaped rash), photosensitivity (skin sensitivity to sunlight), alopecia (hair loss)
Joints/ Muscles	Arthritis-like pain, swollen joints, Raynaud's phenomenon (bluish discoloration of fingers/toes due to vasospasm)
Kidneys	Lupus nephritis (kidney inflammation), proteinuria (protein in urine), hematuria (blood in urine), hypertension (high blood pressure), edema (swelling), risk of renal (kidney) failure
Heart	Pericarditis (inflammation around heart), myocarditis (heart muscle inflammation), murmurs (abnormal heart sounds), chest pain
Lungs	Pleuritis (inflammation around lungs), pleural effusion (fluid around lungs), chest pain, pneumonia (lung infection) risk
Brain 	Fatigue, memory loss, seizures, psychosis (loss of reality), mood disorders
Eyes/Mouth	Dry, red eyes; oral ulcers (sores)
Blood	Anemia (low red blood cells), thrombocytopenia (low platelets), leukopenia (low white blood cells), risk of clots, antiphospholipid syndrome (blood clotting disorder)
General Symptoms	Fever, severe fatigue

RISK FACTORS & TRIGGERS

Gender/Age: Women of childbearing age (especially Black, Latina, Asian women).

Hormonal: Menstruation (monthly period) onset, pregnancy, postpartum (after childbirth) worsen symptoms.

Genetics: Family history.

Environment: Sun exposure, medications, infections.

KEY LABS FOR DIAGNOSIS & MONITORING

ANA (Antinuclear Antibodies): Positive indicates autoantibodies (antibodies attacking own body) against cell nuclei.

Anti-double-stranded DNA (Anti-dsDNA): Specific to lupus, correlates with severity.

Anti-Smith (Anti-Sm): Highly specific to lupus.

ESR (Erythrocyte Sedimentation Rate) & CRP (C-Reactive Protein): Elevated during inflammation and flares.

Complement C3 & C4: Low levels during active disease (used up by inflammation).

CBC (Complete Blood Count): Checks anemia, leukopenia, thrombocytopenia.

Metabolic Panel: Monitor kidney function (BUN - Blood Urea Nitrogen, creatinine).

Urinalysis: Proteinuria indicates lupus nephritis.

MEDICATIONS & TREATMENTS

Goals: Reduce inflammation, prevent flares, protect organs.

1. CORTICOSTEROIDS (PREDNISONE)

Rapid relief from severe inflammation.

Short-term use; taper (gradually reduce) dosage to prevent side effects (infection, osteoporosis (bone weakening), hyperglycemia (high blood sugar)).

2. NSAIDS (NONSTEROIDAL ANTI-INFLAMMATORY DRUGS; IBUPROFEN, NAPROXEN)

Manage joint pain, fever, mild inflammation.

Risks: GI (gastrointestinal) bleeding, ulcers, kidney impairment. Take with food.

3. ANTIMALARIALS (HYDROXYCHLOROQUINE)

Long-term management to prevent flare-ups.

Takes months to show effects.

Risk of retinal damage (eye damage): annual eye exams required.

Smoking decreases effectiveness.

4. IMMUNOSUPPRESSANTS (AZATHIOPRINE, MYCOPHENOLATE MOFETIL)

Severe lupus cases; "steroid-sparing" (reduce steroid usage).

Increase infection risk—avoid live vaccines (weakened live virus vaccines).

5. BIOLOGICS (BELIMUMAB)

Targets B-cell proteins to reduce antibodies and inflammation.

Given by injection or infusion (IV drip); takes up to 6 months for effects.

Assess for mental health risks (depression, suicidal ideation (thoughts of suicide)).



PREGNANCY & LUPUS

Pregnancy increases lupus flares; risk of miscarriage/clots.

Lupus should be controlled (no flare-ups for at least 6 months) before conception (getting pregnant).

Close monitoring needed during pregnancy and postpartum.

RECOGNIZING LUPUS FLARES (MNEMONIC: "FLARE")

F: Fatigue

L: Low-grade fever

A: Achy joints

R: Rashes (butterfly, discoid)

E: Edema (hands, feet swelling)

NURSING MANAGEMENT (MNEMONIC: "LUPUS")

L: LABS

Regular monitoring of ANA, Anti-dsDNA, CBC, ESR, CRP, complement, renal function tests.

U: USE OF MEDICATIONS

Educate about steroids, NSAIDs, antimalarials, immunosuppressants, biologics.

Monitor side effects and compliance (taking medication as prescribed).

P: PREGNANCY PRECAUTIONS

Disease control before conception.

Educate about flare risks during pregnancy and postpartum.

U: UNDERSTANDING FLARES & PREVENTION (MNEMONIC "LESS")

Teach patients to prevent flares by:

L: Lowering stress (emotional/physical).

E: Exercise regularly (maintain joint mobility).

S: Sufficient sleep (>8 hours/night).

S: Sun protection (SPF (Sun Protection Factor) ≥ 30 sunscreen, hats, protective clothing).

S: SIGNS OF FLARES (MNEMONIC: "FLARE")

Educate patients to recognize early symptoms for prompt intervention.



NURSING INTERVENTIONS: KEY POINTS

Tailor (personalize) care to specific affected systems (e.g., kidneys → monitor BUN/Creatinine, edema, weight).

Joint care: pain management, PT (physical therapy) referrals.

Infection prevention: avoid live vaccines (weakened live viruses like MMR, shingles, nasal flu), strict hygiene, regular immunizations.

Psychological support: monitor emotional health, depression signs.



NCLEX HIGH-YIELD POINTS

Butterfly rash (malar rash) is a hallmark symptom.

ANA positive in most lupus cases; Anti-dsDNA & Anti-Sm are confirmatory tests.

Hydroxychloroquine: Requires regular ophthalmologic (eye) exams.

Lupus nephritis (kidney involvement) demands vigilant monitoring.

Pregnancy: Planned conception, control lupus for at least 6 months.

Immunosuppressive therapies increase infection risk—educate thoroughly.